0.1 Alignment building blocks

Like `equation`, the multi-equation environments `gather`, `align`, and `alignat` are designed to produce a structure whose width is the full line width. This means, for example, that one cannot readily add parentheses around the entire structure. But variants `gathered`, `aligned`, and `alignedat` are provided whose total width is the actual width of the contents; thus they can be used as a component in a containing expression. E.g.,

$$\left.\begin{aligned} B' &= -\partial \times E, \\ E' &= \partial \times B - 4\pi j, \end{aligned}\right\} \qquad \text{Maxwell's equations}$$

```
\begin{equation*}
\left.\begin{aligned}
  B'&=-\partial\times E,\\
  E'&=\partial\times B - 4\pi j,
\end{aligned}
\right\}
\qquad \text{Maxwell's equations}
\end{equation*}
```

Like the `array` environment, these `-ed` variants also take an optional `[t]`, `[b]` or the default `[c]` argument to specify vertical positioning. For maximum interoperability, do not insert a space or line break before the option. See also the note on page 2 regarding page breaking within the `-ed` environments.

"Cases" constructions like the following are common in mathematics:

$$P_{r-j} = \begin{cases} 0 & \text{if } r-j \text{ is odd,} \\ r!\,(-1)^{(r-j)/2} & \text{if } r-j \text{ is even.} \end{cases} \tag{1}$$

and in the amsmath package there is a `cases` environment to make them easy to write:

```
P_{r-j}=\begin{cases}
    0&  \text{if $r-j$ is odd},\\
    r!\,(-1)^{(r-j)/2}&  \text{if $r-j$ is even}.
  \end{cases}
```

Notice the use of `\text` (cf. §??) and the nested math formulas. `cases` is set in `\textstyle`. If `\displaystyle` is wanted, it must be requested explicitly; `mathtools` provides a `dcases` environment for this purpose.

The `-ed` and `cases` environments must appear within an enclosing math environment, which can be either in text, between `$...$`, or in any of the display environments.

0.2 Adjusting tag placement

Placing equation numbers can be a rather complex problem in multiline displays. The environments of the amsmath package try hard to avoid overprinting an equation number on the equation contents, if necessary moving the number

down or up to a separate line. Difficulties in accurately calculating the profile of an equation can occasionally result in number movement that doesn't look right. A `\raisetag` command is provided to adjust the vertical position of the current equation number, if it has been shifted away from its normal position. If it is in its normal position the command has no effect whatsoever. In particular, this means that one can't use it to shift the tag of an `equation` environment.

To move a particular number up by six points, write `\raisetag{6pt}`. (At the end of a display, this also shifts up the text following the display.) This kind of adjustment is fine tuning like line breaks and page breaks, and should therefore be left undone until your document is nearly finalized, or you may end up redoing the fine tuning several times to keep up with changing document contents.

0.3 Vertical spacing and page breaks in multiline displays

You can use the `\\[`⟨*dimension*⟩`]` command to get extra vertical space between lines in all the amsmath displayed equation environments, as is usual in LaTeX. Do not type a space between the `\\` and the following `[`; *only* for display environments defined by amsmath the space is interpreted to mean that the bracketed material is part of the document content.

When the amsmath package is in use page breaks between equation lines are normally disallowed; the philosophy is that page breaks in such material should receive individual attention from the author. To get an individual page break inside a particular displayed equation, a `\displaybreak` command is provided. `\displaybreak` is best placed immediately before the `\\` where it is to take effect. Like LaTeX's `\pagebreak`, `\displaybreak` takes an optional argument between 0 and 4 denoting the desirability of the pagebreak. `\displaybreak[0]` means "it is permissible to break here" without encouraging a break; `\displaybreak` with no optional argument is the same as `\displaybreak[4]` and forces a break.

If you prefer a strategy of letting page breaks fall where they may, even in the middle of a multiline equation, then you might put `\allowdisplaybreaks[1]` in the preamble of your document. An optional argument 1–4 can be used for finer control: `[1]` means allow page breaks, but avoid them as much as possible; values of 2,3,4 mean increasing permissiveness. When display breaks are enabled with `\allowdisplaybreaks`, the `\\*` command can be used to prohibit a pagebreak after a given line, as usual.

Note: Certain equation environments wrap their contents in an unbreakable box, with the consequence that neither `\displaybreak` nor `\allowdisplaybreaks` will have any effect on them. These include `split`, `aligned`, `gathered`, and `alignedat`.

0.4 Interrupting a display

The command `\intertext` is used for a short interjection of one or two lines of text in the middle of a multiple-line display structure (see also the `\text` command in §**??**). Its salient feature is preservation of the alignment, which would not happen if you simply ended the display and then started it up again

afterwards. `\intertext` may only appear right after a `\\` or `\\*` command. Notice the position of the word "and" in this example.

$$A_1 = N_0(\lambda;\Omega') - \phi(\lambda;\Omega'), \tag{2}$$

$$A_2 = \phi(\lambda;\Omega') - \phi(\lambda;\Omega), \tag{3}$$

and

$$A_3 = \mathcal{N}(\lambda;\omega). \tag{4}$$

```
\begin{align}
  A_1&=N_0(\lambda;\Omega')-\phi(\lambda;\Omega'),\\
  A_2&=\phi(\lambda;\Omega')-\phi(\lambda;\Omega),\\
\intertext{and}
  A_3&=\mathcal{N}(\lambda;\omega).
\end{align}
```

The `mathtools` package provides a command `\shortintertext` that is intended for use when the interjected text is only a few words; it uses less vertical space than `\intertext`. This is most effective when equation numbers are on the right.

0.5 Equation numbering

0.5.1 Numbering hierarchy

In LaTeX if you wanted to have equations numbered within sections—that is, have equation numbers (1.1), (1.2), ..., (2.1), (2.2), ..., in sections 1, 2, and so forth—you could redefine `\theequation` as suggested in the LaTeX manual [**?**, §6.3, §C.8.4]:

```
\renewcommand{\theequation}{\thesection.\arabic{equation}}
```

This works pretty well, except that the equation counter won't be reset to zero at the beginning of a new section or chapter, unless you do it yourself using `\setcounter`. To make this a little more convenient, the amsmath package provides a command `\numberwithin`. To have equation numbering tied to section numbering, with automatic reset of the equation counter, write

```
\numberwithin{equation}{section}
```

As its name implies, the `\numberwithin` command can be applied to any counter, not just the `equation` counter.

0.5.2 Cross references to equation numbers

To make cross-references to equations easier, an `\eqref` command is provided. This automatically supplies the parentheses around the equation number. I.e., if `\ref{abc}` produces 3.2 then `\eqref{abc}` produces (3.2). The parentheses around an `\eqref` equation number will be set in upright type regardless of type style of the context.

0.5.3 Subordinate numbering sequences

The amsmath package provides also a wrapper environment, `subequations`, to make it easy to number equations in a particular `\align` or similar group with a subordinate numbering scheme. For example

```
\begin{subequations}
...
\end{subequations}
```

causes all numbered equations within that part of the document to be numbered (4.9a) (4.9b) (4.9c) . . . , if the preceding numbered equation was (4.8). A `\label` command immediately after `\begin{subequations}` will produce a `\ref` of the parent number 4.9, not 4.9a. The counters used by the subequations environment are `parentequation` and `equation` and `\addtocounter`, `\setcounter`, `\value`, etc., can be applied as usual to those counter names. To get anything other than lowercase letters for the subordinate numbers, use standard LaTeX methods for changing numbering style [**?**, §6.3, §C.8.4]. For example, redefining `\theequation` as follows will produce roman numerals.

```
\begin{subequations}
\renewcommand{\theequation}{\theparentequation \roman{equation}}
...
```

0.5.4 Numbering style

The default equation number is set in `\normalfont`. This means that in bold section headings, bold is suppressed; a workaround for this is to use `(\ref{...})` rather than `\eqref{...}`.

If smaller type is specified for a numbered display, the size of the equation number will also be small. The default size can be ensured throughout a document by applying this patch in the preamble:

```
\makeatletter
\renewcommand{\maketag@@@}[1]{\hbox{\m@th\normalsize\normalfont#1}}%
\makeatother
```

(This modification may be included in a future version of amsmath.)